

- 8 (a) Fig. 8.1 is a diagram of a sensory neurone.

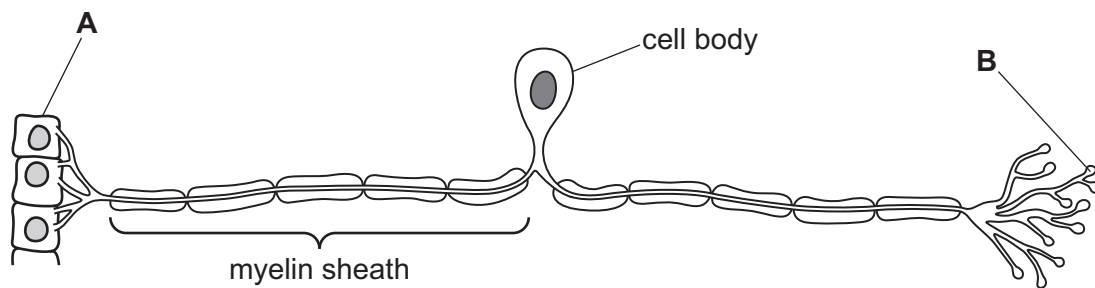


Fig. 8.1

- (i) Name the type of cell represented by cell **A**.

..... [1]

- (ii) Name a type of cell that forms a synapse with structure **B**.

..... [1]

- (iii) Name the cells that form the myelin sheath.

..... [1]

- (b) State **and** explain the differences in the transmission of impulses between myelinated and unmyelinated neurones.

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..... [4]



- (c) Opioids are therapeutic drugs that are used to relieve pain. Morphine is an example of an opioid. It affects the functioning of synapses.

Fig. 8.2 is a diagram of a cholinergic synapse.

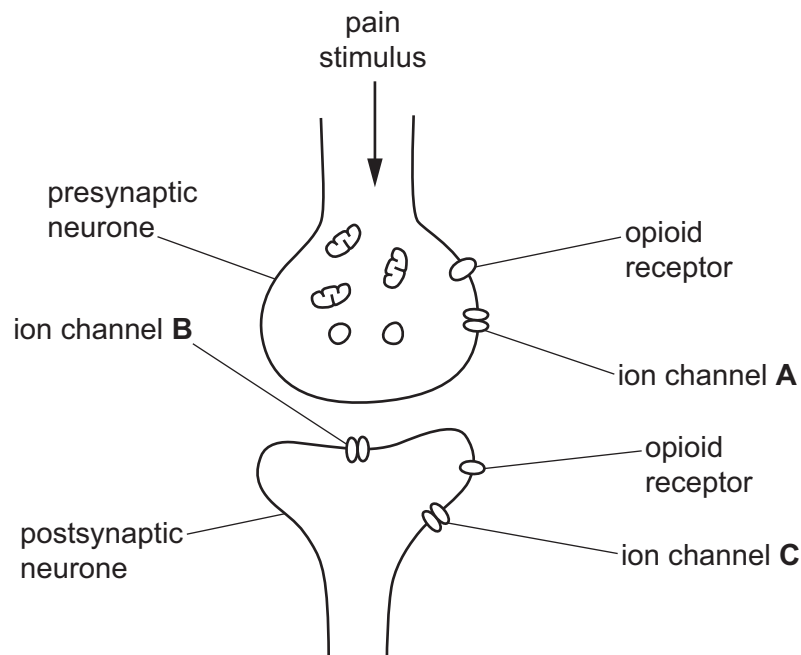


Fig. 8.2

- (i) With reference to Fig. 8.2, describe the differences between ion channel **A** and ion channel **B**.

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..... [2]



- (ii) Morphine binds to the opioid receptor in the presynaptic membrane. This stops the ion channel **A** from opening.

Suggest **and** describe the effects of the binding of morphine to this opioid receptor on the functioning of the cholinergic synapse.

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- (iii) Morphine also binds to the opioid receptor in the postsynaptic membrane. This results in the opening of ion channel **C**, allowing potassium ions to diffuse out of the cell.

Suggest how this would affect the synapse.

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..... [2]

[Total: 15]